Rant or Rave: Variation over Time in the Language of Online Reviews

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1. Introduction

Online reviews have become a vital resource for consumer decision-making. As e-commerce and social media rapidly grow, consumers increasingly depend on others' experiences to reduce purchase risks and fill information gaps. Surveys show that over 90% of consumers read online reviews before making a purchase, with most dedicating at least a minute to them. This demonstrates the growing impact of reviews on shaping buying choices. For platforms and businesses, reviews not only influence sales and reputation but also play a crucial role in recommendation algorithms and market competitiveness.

However, the language used in reviews is not fixed. Over time, the way people express themselves in online reviews has changed significantly. For example, Ziser, Webber & Cohen (2023), in their study "Rant or Rave: Variation over Time in the Language of Online Reviews," found that on platforms like Yelp, IMDb, and Amazon, reviews have become shorter, more polarized, and increasingly depend on frequent sentiment words.

Studying these changes is essential for several reasons. From the consumer point of view, review length, sentiment, and lexical diversity impact perceptions of informativeness and trustworthiness. Shorter or overly biased reviews may lessen their usefulness or even mislead readers. From the perspective of businesses and platforms, review quality directly influences user experience and platform credibility; understanding how review language has evolved can help develop more effective recommendation and ranking systems. Academically, online review data offers valuable insights into language development and human–computer interaction.

Prompted by these reasons, this study aims to systematically analyze the linguistic features of reviews over time, focusing on Amazon's Electronics category from 2008 to 2023. Our goal is to partly replicate and expand on the work of Ziser, Webber & Cohen (2023), using more recent data to examine ongoing trends. Specifically, we seek to answer these research questions:

• RQ1: How has the way sentiment is expressed in online reviews changed from 2008 to 2023?

• RQ2: What are the long-term trends in review length and polarization?

• RQ3: Has the use of extreme sentiment words become more concentrated?

• RQ4: How do positive and negative reviews differ in these patterns of change?

2. Data Collection

The data used in this study comes from the Amazon Product Reviews 2023 dataset, which was compiled and publicly released by the McAuley team at the University of California, San Diego (UCSD). This dataset spans multiple categories from 1996 to 2023 and includes over 500 million reviews, making it one of the largest publicly available review datasets. We focused on the Electronics category and looked at four specific years: 2008, 2013, 2018, and 2023.

We chose the four time points of 2008, 2013, 2018, and 2023 because, compared to yearly analysis, these key years allow us to capture major linguistic shifts while keeping computational costs manageable. The period from 2008 to 2023 covers 15 years, providing enough time to observe long-term changes in review languages. Selecting data points roughly every five years allows us to capture the phased nature of linguistic shifts.

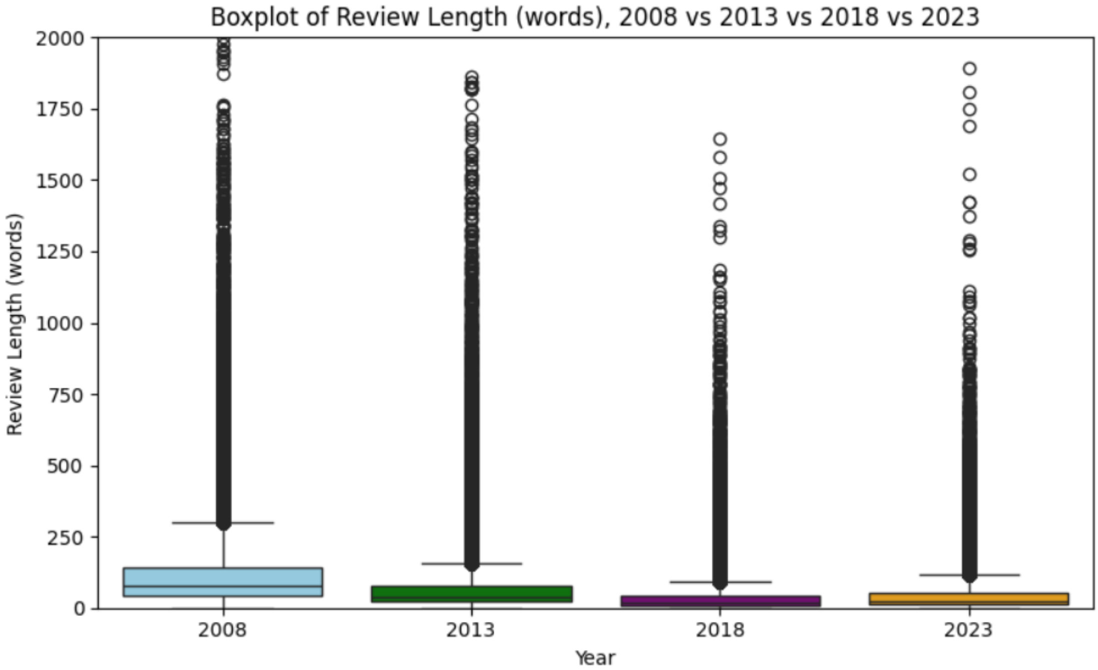
Due to the large volume of raw data, processing all reviews directly was unfeasible because of storage and computational constraints. Therefore, we employed random sampling to reduce the dataset. Specifically, we used the streaming mode from the Hugging Face datasets library to read reviews sequentially, filtered data by review timestamps to target specific years, and within each year, randomly sampled 100,000 reviews using reservoir sampling to ensure representativeness. During data cleaning, we removed corrupted and empty rows and standardized timestamp formats. We ultimately retained the following core fields:

- rating (star rating, 1–5 stars)
- title (review title)
- text (review body)
- user_id (user identifier)
- asin (product identifier)
- timestamp (review posting time)

This resulted in approximately 400,000 reviews (4 years × 100,000 reviews), serving as the foundation for further analysis.

3. Data Overview

The star rating distribution shows clear polarization over time. In 2008, 53.0% of reviews were 5-star and 10.6% were 1-star. By 2013, 5-star reviews peaked at 58.4% while 1-star reviews slightly declined to 9.1%. In 2018, 5-star reviews rose sharply to 65.0% and 1-star reviews increased to 12.1%. By 2023, 5-star reviews slightly declined to 62.3%, but 1-star reviews reached their highest level at 15.3%, indicating growing polarization at both extremes while mid-range ratings (2–4 stars) steadily declined overall.



Review length declined dramatically over the 15-year period. The median review length dropped from 78 words in 2008 to 40 words in 2013, reaching a low of just 18 words in 2018, before slightly recovering to 25 words in 2023. Mean length followed the same pattern: 121.7 words (2008) → 71.2 (2013) → 36.3 (2018) → 44.0 (2023). This consistent shortening suggests that reviews have become increasingly brief and less detailed over time.

4. Methods

4.1 Sentiment Analysis (VADER)

I used the VADER (Valence Aware Dictionary for Sentiment Reasoning) lexicon-based method to analyze each review individually. For each review, I calculated and stored sentiment scores, including the compound (overall polarity), pos (proportion of positive words), and neg (proportion of negative words). This provided sentiment scores for all reviews over four years, which I used to identify overall trends. Later, I grouped the reviews based on star ratings (1–2 stars as Negative, 4–5 stars as Positive) to compare positive and negative reviews.

4.2 Language diversity and comprehensiveness

To analyze the use of extreme sentiment words, we extracted high-intensity terms from the VADER lexicon (valence $\geqslant$ 3 or $\leqslant$ -3) and tokenized review texts for word-level matching. We computed the following metrics: (1) the proportion of extreme sentiment words among all tokens; (2) their proportion among sentiment-bearing tokens; (3) the diversity of sentiment word types and the coverage of the top 1% most frequent sentiment words. These statistics were aggregated for each year (2008, 2013, 2018, 2023) and further broken down by Positive and Negative reviews. Additionally, we visualized the overall (ungrouped) share of extreme words across years and grouped (positive/negative) extreme word share across years.

4.3 Review Length and One-Sidedness Analysis

We analyzed review length and one-sidedness using a lexicon-based approach. Using the VADER lexicon, we built sets of positive and negative words, including common negation markers (e.g., not, never, don't). When a negation appeared within a short window before a sentiment word, we reversed its polarity. Each review was tokenized to count positive and negative words, helping us identify one-sided reviews—either positive reviews with only positive words or negative reviews with only negative words. Review length was measured in words, and averages were calculated for each year and sentiment category (Positive/Negative). Finally, we plotted yearly trends of average length, one-sidedness, and the concentration of extreme sentiment words to observe changes in review language from 2008 to 2023.

4.4 Illustrative Examples of Extreme Sentiment Words

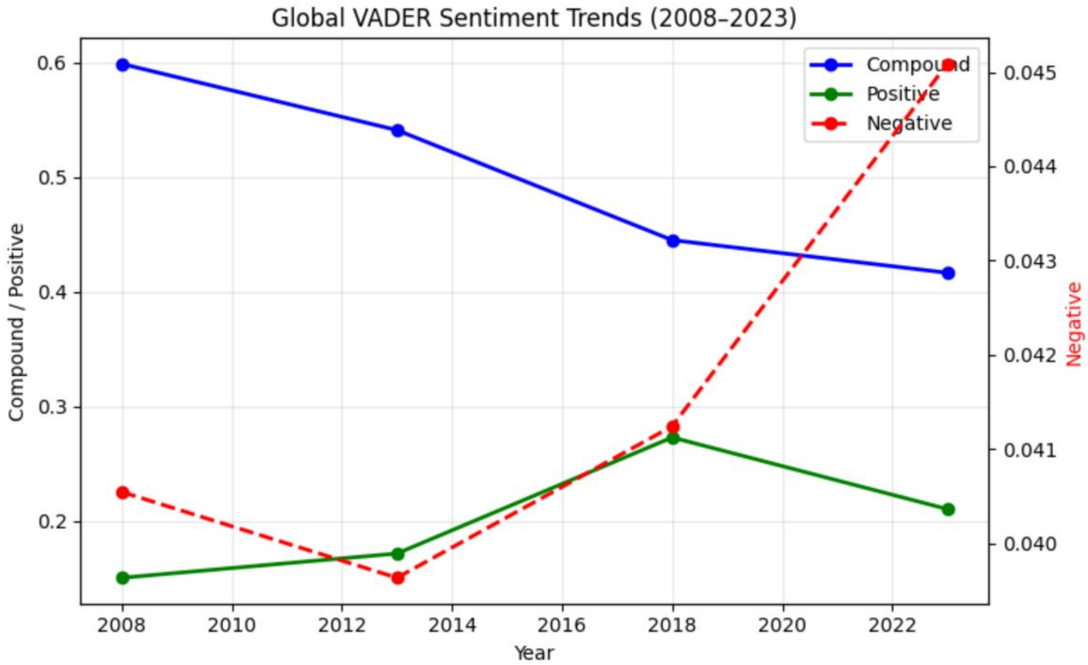
To supplement the overall statistics, we selected sample reviews containing extreme sentiment words. We then picked the shortest reviews with at least one such word, limiting the sample to five examples for each class (Positive vs. Negative) and year (2008, 2023). This demonstrates how even very brief reviews can still express strong emotional polarity through the use of extreme sentiment words.

4.5 BERT vs. VADER Comparison

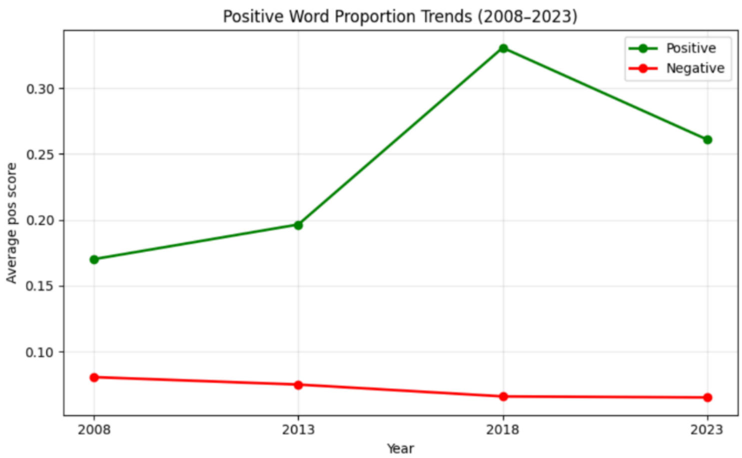
To complement the rule-based VADER approach, we applied a transformer-based sentiment model (nlptown/bert-base-multilingual-uncased-sentiment, accessed via Hugging Face Transformers) to a stratified random sample of 500 reviews per year (2,000 reviews in total). BERT predicts a star rating from 1 to 5, which we mapped to three classes: Positive (4–5 stars), Neutral (3 stars), and Negative (1–2 stars). VADER compound scores were mapped using standard thresholds ($\geq 0.05 \rightarrow$ Positive; $\leq -0.05 \rightarrow$ Negative; otherwise Neutral). We then measured the label agreement rate between the two models overall, by year, and by rating class, and visualized disagreements using a confusion matrix.

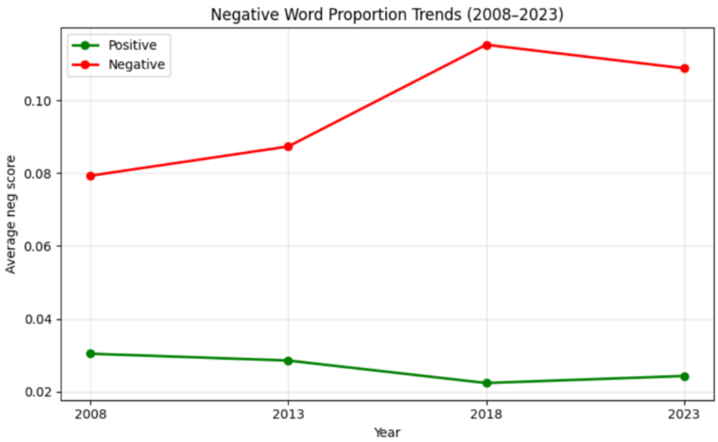
5. Results

5.1 Sentiment Analysis (VADER)



The global VADER compound score declined steadily from 0.599 in 2008 to 0.541 in 2013, 0.445 in 2018, and 0.417 in 2023, indicating consistently less positive overall sentiment. The share of positive words increased until 2018 (peaking at 0.273) before declining to 0.211 in 2023, while the proportion of negative words rose from 0.041 (2008) to 0.045 (2023).



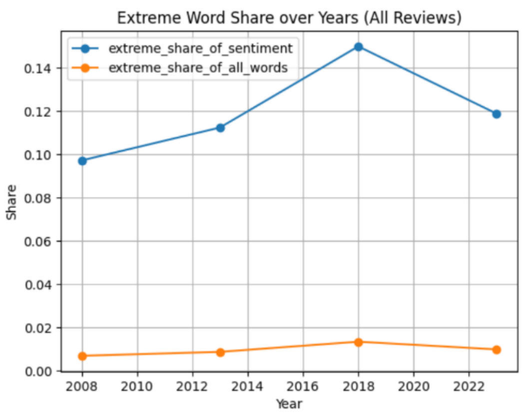


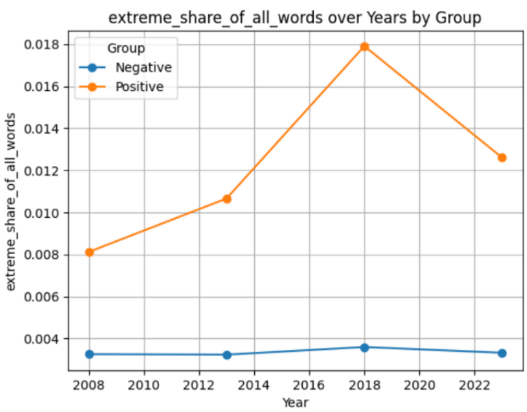
When split by rating class, positive reviews showed a compound score decline from 0.737 (2008) to 0.583 (2023), while negative reviews shifted from near-neutral (0.050 in 2008) to clearly negative (−0.097 in 2023), suggesting that negative reviews have become more expressively critical. The proportion of negative reviews also grew, from 16.4% of all reviews in 2008 to 20.8% in 2023.

Overall, from 2008 to 2023, the sentiment of Amazon Electronics reviews has gradually shifted toward negativity. The consistent decline in compound scores indicates that the overall tone of reviews is less positive than in earlier years. Notably, around 2018 marked a significant turning point: before this year, the share of positive words steadily increased while negative expressions remained relatively low; however, after 2018, positive sentiment dropped sharply, and negative sentiment rose quickly.

This suggests that consumers have become generally less satisfied with electronic products, which may be explained by rising standards and stricter expectations as the electronics industry has expanded.

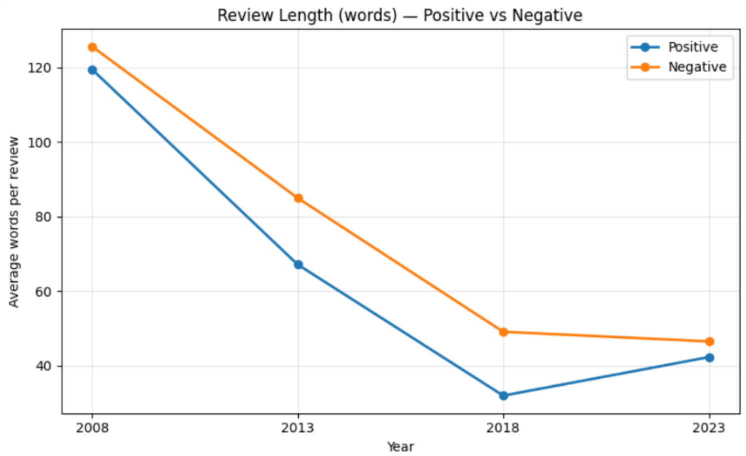
5.2 Language diversity and comprehensiveness



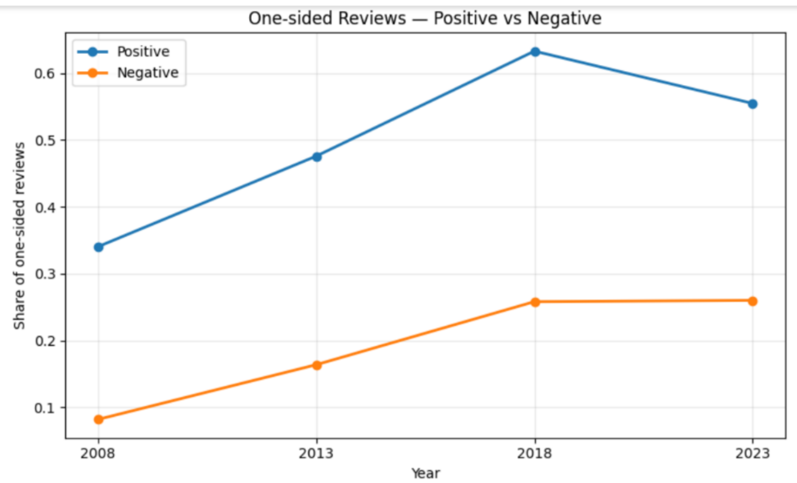


Our analysis of extreme sentiment word usage shows that over time, reviews have become more focused on a small set of very common extreme words, especially in positive reviews (e.g., great, love, best). This suggests a decrease in language variety: instead of using a wide range of descriptive words, reviewers are increasingly relying on a few repetitive, emotionally charged terms.

5.3 Review Length and One-Sidedness Analysis



Between 2008 and 2023, both positive and negative reviews became much shorter. Positive reviews fell from an average of 119.4 words in 2008 to just 32.0 words in 2018, recovering slightly to 42.3 words in 2023. Negative reviews dropped from 125.6 words in 2008 to 49.1 words in 2018 and 46.5 words in 2023. By 2023, the average lengths of positive and negative reviews had converged, reflecting a general trend toward brevity regardless of sentiment.



One-sidedness increased substantially for both groups. Among positive reviews, the share of one-sided reviews rose from 34.0% in 2008 to a peak of 63.3% in 2018, then slightly declined to 55.5% in 2023. Negative reviews showed a similar but less pronounced trend, rising from 8.2% in 2008 to 25.8% in 2018 and remaining stable at 26.0% in 2023. These findings suggest that reviewers increasingly express a single emotional perspective rather than a balanced evaluation, reducing the overall informativeness of reviews.

5.4 Illustrative Examples of Extreme Sentiment Words

To illustrate the use of extreme sentiment words, we selected the shortest reviews containing at least one extreme term for 2008 and 2023. In 2008, the shortest positive reviews already contained words like great and awesome (e.g., "Great product :)", "Works great!"), while negative reviews, though short, still provided some context (e.g., "Dead in 30 min. Sent back for full refund."). By 2023, positive reviews had collapsed to single words such as "AWESOME" or "Awesome," repeated across many entries, while negative reviews similarly reduced to bare phrases like "Worst product" or "Worst webcam." This contrast vividly illustrates how extreme sentiment words have become the entire review rather than part of a longer evaluation.

5.5 BERT vs. VADER Comparison

The overall VADER–BERT label agreement was 72.7% across 2,000 sampled reviews (500 per year). Agreement improved over time: 67.8% (2008) → 73.0% (2013) → 75.8% (2018) → 74.0% (2023), suggesting that shorter, more direct modern reviews are somewhat easier to classify consistently. Agreement varied strongly by rating class: positive reviews achieved 82.1% agreement, while negative reviews reached only 47.5% and neutral reviews just 33.8%. At the class level, BERT correctly identified 89.6% of VADER-labeled positive reviews, but only 45.3% of negative and 11.3% of neutral ones. This confirms that VADER underperforms on negative and ambiguous texts, where BERT's contextual understanding provides a clear advantage.

6. Conclusion

Between 2008 and 2023, Amazon Electronics reviews underwent a clear transformation: reviews became notably shorter (median length from 78 to 25 words), more one-sided (positive one-sidedness from 34.0% to 55.5%), and more uniform in language, with extreme sentiment words like great, awesome, worst, and dead dominating the discussion. Meanwhile, overall sentiment shifted from predominantly positive (compound score 0.599) to noticeably less positive (0.417), with negative reviews growing from 16.4% to 20.8% of all reviews.

A complementary VADER–BERT comparison on 2,000 sampled reviews revealed an overall agreement of 72.7%, with the highest consistency for positive reviews (82.1%)

and the lowest for neutral ones (33.8%). This highlights a key limitation of rule-based approaches: BERT's contextual understanding makes it substantially better suited for negative and ambiguous texts — an increasingly important consideration as reviews become shorter and more opaque.

7. Critique and Reflection

This study has several limitations. First, we sampled only four years (2008, 2013, 2018, and 2023) instead of analyzing the entire time series. Although this approach provides a manageable scope and still captures long-term trends, it might miss fluctuations or turning points in intermediate years. Second, our analysis focused solely on the Amazon Electronics category, which could limit the applicability of the findings to other domains or platforms. Future research could expand the temporal scope, include multiple product domains, and utilize more advanced models (e.g., deep learning–based sentiment analysis) to provide a more detailed understanding of how online review language evolves.

8. Literature

Ziser, Y., Webber, B., & Cohen, S. (2023). *Rant or rave: Variation over time in the language of online reviews*. Transactions of the Association for Computational Linguistics, 11, 1–17.